

Trainings ▾

Preparatory Steps

⚠ WARNING ⚠

To reduce the possibility of personal injury, avoid direct contact of hot oil with your skin.

⚠ WARNING ⚠

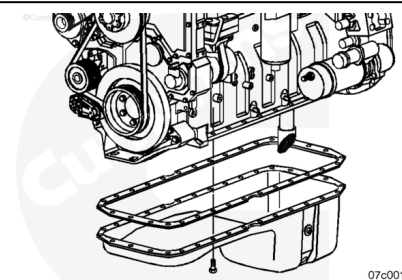
Some state and federal agencies have determined that used engine oil can be carcinogenic and cause reproductive toxicity. Avoid inhalation of vapors, ingestion, and prolonged contact with used engine oil. If not reused, dispose of in accordance with local environmental regulations.

- Drain the lubricating oil. Refer to Procedure 007-037 (Lubricating Oil System) in Section 7. (</qs3/pubsys2/xml/en/procedures/10/10-007-037-tr.html>)

Remove

Remove the oil pan mounting capscrews, oil pan, and gasket.

Discard the oil pan gasket.



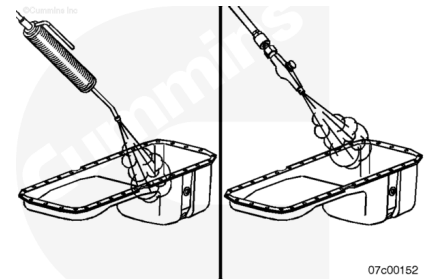
Clean and Inspect for Reuse

⚠ WARNING ⚠

When using a steam cleaner, wear safety glasses or a face shield, as well as protective clothing. Hot steam can cause serious personal injury.

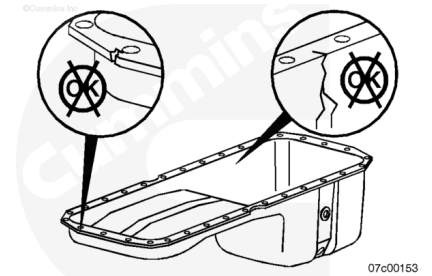
⚠ WARNING ⚠

Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause bodily injury.



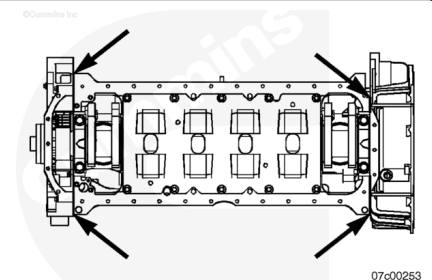
Steam clean and dry with compressed air.

Inspect the inside and the outside of the oil pan for cracks.
Replace the oil pan if cracked or broken.



Install

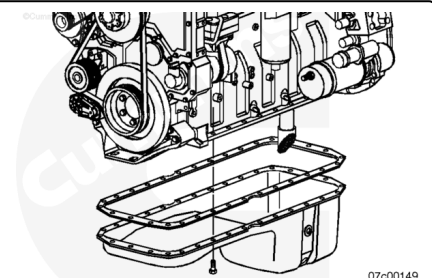
Apply a 1.97 mm [1/8 in] bead of silicon sealant, Part Number 3164067, to each of the four T-joints where the front gear housing and flywheel housing gaskets intersect the oil pan gasket.



Install a new oil pan gasket.

Install the oil pan onto the cylinder block. Install one of the 13 mm oil pan mounting capscrews, hand-tight, on each side of the oil pan, halfway between the front and the rear of the oil pan.

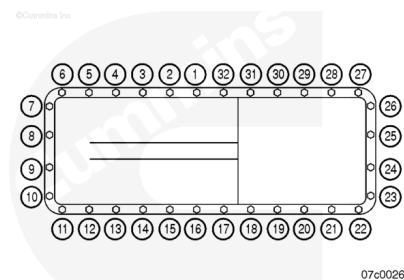
Finger tighten all of the mounting capscrews.



Tighten the thirty-two 13 mm capscrews in the sequence shown.

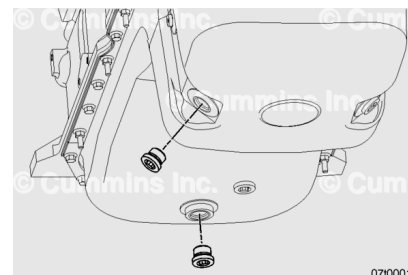
At the end of the sequence, loosen and tighten capscrew number 1 again.

Torque Value: 55 n•m [41 ft-lb]



Install the oil drain plug(s).

Torque Value: 55 n•m [41 ft-lb]



Finishing Steps

Fill the lubricating oil pan. Refer to Procedure 007-037(Lubricating Oil System) in Section 7. (</qs3/pubsys2/xml/en/procedures/10/10-007-037-tr.html>)

Operate the engine until the water temperature reaches 82°C [180°F]. Check for leaks.